Dementia Risk Profiling from Clinical Narratives Using Small LLMs

Sean Yu
Independent Researcher
seanjasparyu@gmail.com

April 10, 2026

Abstract

This study presents a proof of concept for an automated pipeline designed to extract dementia risk factors from unstructured clinical narratives and generate computable risk scores. Utilizing patient summaries from the PMC-Patients dataset, the research evaluates a workflow employing a Large Language Model (LLM) for feature extraction alongside text embeddings and an XGBoost classifier for stratification. Results indicate that high-dimensional text embeddings provide the dominant predictive signal, with the model using Demographics, Embeddings, and LLM features achieving an AUC of 0.875 and a peak accuracy of 0.814. This feasibility study confirms that generative AI can effectively bridge the gap between 'locked' narrative data and structured clinical decision support, providing a scalable framework for identifying at-risk populations within electronic health records.

1 Introduction

As dementia cases are projected to reach 153 million by 2050, the ability to address the 40% of cases linked to modifiable risk factors has become a global health priority. However, critical patient data regarding these risks is currently "locked" within unstructured clinical narratives that standard electronic health records fail to capture accurately. While Large Language Models (LLMs) offer a potential key to extracting this information, their clinical utility is often hindered by concerns over data privacy, computational costs, and the risk of hallucinations. This disconnect represents a significant missed opportunity for population risk stratification and real-time clinical decision support.

To address these barriers, this project presents a proof of concept evaluating whether a LLM can reliably extract risk factors to populate a binary classifier, ultimately generating a probability-based risk score for dementia. By utilizing publicly available PMC-Patients data, the study establishes an end-to-end workflow designed to run on consumer-grade GPUs, bypassing the privacy risks associated with commercial APIs. The primary objective is to quantify the "added value" of generative AI by comparing its performance against simpler baselines, such as regex and embeddings, while identifying specific failure points through manual validation. Ultimately, this work seeks to demonstrate a feasible, transparent pipeline for extracting computable clinical insights from free-text narratives.

2 Background Information, Related Work and Dataset

2.1 Clinical Risk Factors for Dementia

The Lancet Commission on Dementia prevention, intervention, and care [Liv+24] outline 14 major risk factors that they claim if are properly managed could reduce the total worldwide dementia cases by 45%. Table 1 summarizes the risk factors. Based on these factors, the following terms/diseases were searched in the clinical text using the LLM: hypertension, brain injury, diabetes, depression, smoking, obesity, afib, stroke, alcohol.

Life Stage	Risk Factors
Early Life (Ages 0-18)	Less education
Midlife (Ages 18-65)	Hearing loss, High LDL cholesterol, Depression, Traumatic brain injury, Physical inactivity, Diabetes, Smoking, Hypertension,, Obesity, Excessive alcohol
Later Life (Ages 65+)	Social isolation, Air pollution, Visual loss

Table 1: Lancet Commision Dementia Risk Factors

2.2 Related Work in Clinical Text Extraction

With the advent of LLMs, text extraction has transitioned from rigid, rule-based patterns to fluid, context-aware reasoning that can interpret nuance and intent without the need for manual feature engineering. Rule based methods, such as clinical Text Analysis and Knowledge Extraction System (cTakes) [Sav+10] and Clinical Language Annotation, Modeling and Processing (CLAMP) [Soy+18], are typically pipeline-based meaning they rely on separating text and identifying individual tokens, parts of speech, and clinical concepts through a series of modular stages, each of which must be meticulously tuned to ensure accuracy. While interpretable, they are relatively labor intensive due to the fine-tuning required.

While rule-based pipelines excel in transparency, they often struggle with the linguistic variability of clinical prose. Bidirectional Encoder Representations from Transfomer (BERT) models such as BioBERT [Lee+19] and ClinicalBERT [HAR19] mitigate this by using deep learning to capture word meaning based on surrounding context rather than rigid patterns. However, while these models offer superior accuracy, their performance is contingent on access to high-performance GPU clusters and the availability of large, domain-specific training datasets.

The most recent shift in the field is the adoption of LLMs, which move beyond the need for task-specific fine-tuning. By utilizing zero-shot or few-shot prompting, LLMs can perform complex extraction tasks immediately upon deployment, interpreting natural language instructions rather than relying on rigid rules or vast annotated datasets. This flexibility allows for rapid iteration across different clinical domains without the 'data bottleneck' inherent to previous architectures.

2.3 The PMC-Patients Dataset

The PMC-Patients dataset [Zha+23] comprises approximately 167,000 patient summaries derived from case reports within PubMed Central. Each entry provides a rich free-text narrative alongside structured demographic metadata (age and gender) and relational annotations for patient-article and patient-patient similarity. As an open-access resource requiring no formal data use agreement, it has become a cornerstone for benchmarking retrieval-based clinical decision support systems and, crucially, provides a diverse corpus for evaluating clinical text extraction models.

3 Experiment and Results

Due to computational constraints, the study was limited to a sample of 700 patients. This experiment utilized an evenly distributed dataset consisting of 350 patients with a dementia diagnosis and 350 control patients. The control group was selected by identifying cases containing keywords unrelated to dementia risk factors, such as pneumonia, hip fracture, and knee replacement. The full details can be found in Appendix A.

To evaluate the effectiveness of LLMs for clinical data extraction, a two stage approach was employed. First, the LLM was tasked with identifying specific features from a patient summary. Second, the summary was converted into text embeddings using a sentence transformer to facilitate downstream analysis. In this experiment the transformer "all-MiniLM-L6-v2" was used because it offers a favorable trade-off between semantic representation quality and computational efficiency. These components were then evaluated through an ablation study using an XGBoost classifier. Details regarding the model configurations and hyperparameters are located in Appendix B. This process tested the predictive power of various feature sets, including demographics such as age and gender, text embeddings, and extracted features. The experiments were performed using the LLM "gemma-2b-it" the results are presented in Table 2.

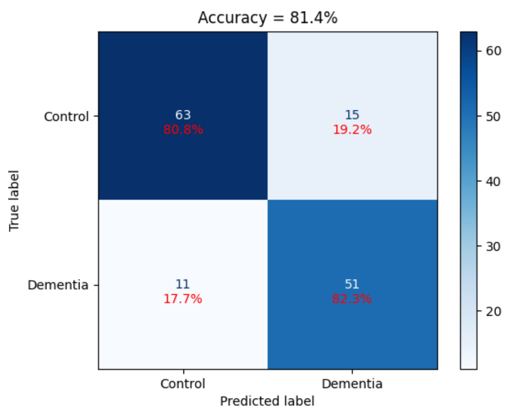
The bold values indicate the highest scores. The results demonstrate that incorporating text representations is critical for dementia prediction. Models using text embeddings substantially outperformed those relying solely on demographics, regardless of whether additional structured features were included. Adding LLM-extracted or regex-extracted risk

Dataset	AUC	F1	Recall	Accuracy
Demographics only	0.551	0.549	0.629	0.543
Demographics + Text embeddings	0.880	0.794	**0.839**	0.807
Demographics + Text Embeddings + LLM features	0.875	**0.797**	0.823	**0.814**
Demographics + Text Embeddings + Regex features	**0.894**	0.769	0.806	0.786
Demographics + LLM features	0.659	0.619	0.774	0.579
Demographics + regex features	0.640	0.596	0.677	0.593

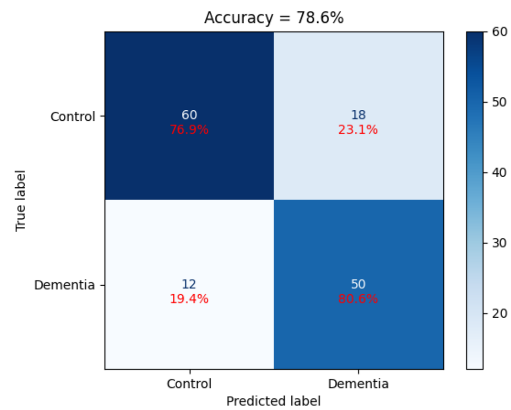
Table 2: Classification results for dementia prediction (N=700)

factors yielded comparable performance to embeddings alone, with all text-based models achieving AUC between 0.875 and 0.894. This suggests that dense text embeddings capture the relevant clinical signal effectively, and that explicit risk factor extraction, whether via LLM or regex, could provide minimal marginal benefit in this dataset. Notably, LLM-extracted features alone (without embeddings) achieved AUC 0.659, significantly outperforming demographics alone (AUC 0.551), confirming that the extracted features are clinically meaningful even when used in isolation.

The figure 1 show the confusion matrices of the models with trained with (Demographics + Text Embeddings + LLM features) and (Demographics + Text Embeddings + Regex features).



(a) Demographic + Text Embedding + LLM extracted features



(b) Demographic + Text Embedding + Regex extracted features

Figure 1: Confusion Matrices

The confusion matrices show that LLM-based feature extraction outperforms the Regex approach with an overall accuracy of 81.4%, demonstrating a superior ability to identify both Dementia and Control cases accurately. By capturing nuanced semantic context rather than relying on rigid pattern matching, the LLM features successfully reduced both false positives and false negatives.

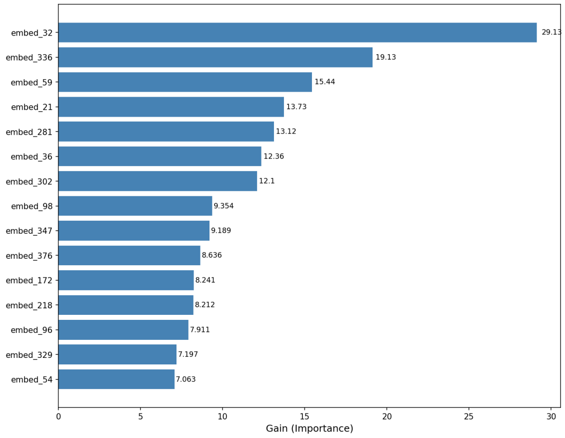
The figure 2 contains plots with the top 15 feature importances as given by the XGB models.

Both feature importance plots demonstrate that embedding features exert a dominant influence on the model's decisions. This confirms that high-dimensional semantic representations are more effective than rigid, pattern-based features at capturing the subtle linguistic markers of cognitive decline. This could also imply that the extracted features are possibly missing key information which gives the text embedding an advantage.

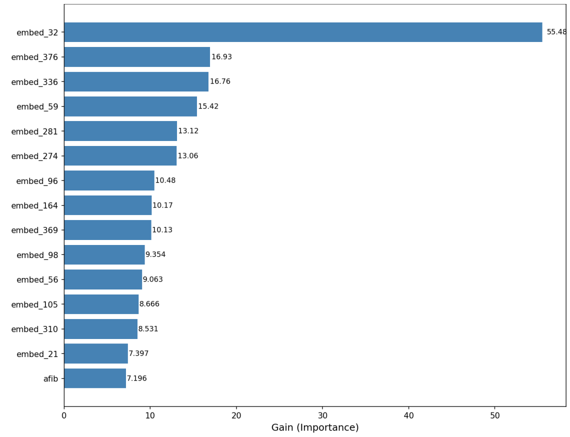
4 Discussion and Conclusion

The results demonstrate a clear hierarchy in feature efficacy for dementia prediction. While demographics alone provided a negligible predictive signal, the integration of dense text embeddings elevated the predictive performance. This suggests that high-dimensional semantic representations are uniquely capable of capturing the subtle, non-linear linguistic markers of cognitive decline that traditional structured data misses.

A key finding is the "performance plateau" observed when adding explicit LLM or Regex-extracted features to embedding-based models. While these extracted features are clinically grounded and outperformed demographics in isolation, they offered minimal marginal utility when paired with embeddings. This indicates that while embeddings are



(a) Demographic + Text Embedding + LLM extracted features



(b) Demographic + Text Embedding + Regex extracted features

Figure 2: XGB feature importance

highly effective predictors, there remains a trade-off between the predictive power of "black-box" embeddings and the interpretability of explicit clinical markers.

It is worth noting that this study utilized general-purpose LLM and text embedding models. While these achieved strong results, the existence of models fine-tuned specifically for clinical use provides a clear path for further optimization. Utilizing domain-specific architectures could potentially refine the feature extraction process and further improve the classification of subtle pathological linguistic markers

References

[Sav+10] Guergana K Savova et al. "Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications". en. In: *J. Am. Med. Inform. Assoc.* 17.5 (Sept. 2010), pp. 507–513.

[Soy+18] Ergin Soysal et al. "CLAMP - a toolkit for efficiently building customized clinical natural language processing pipelines". en. In: *J. Am. Med. Inform. Assoc.* 25.3 (Mar. 2018), pp. 331–336.

[HAR19] Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. "ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission". In: *arXiv:1904.05342* (2019).

[Lee+19] Jinhyuk Lee et al. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining". In: *Bioinformatics* 36.4 (Sept. 2019). Ed. by Jonathan Wren, pp. 1234–1240. ISSN: 1367-4811. DOI: 10.1093/bioinformatics/btz682. URL: http://dx.doi.org/10.1093/bioinformatics/btz682.

[Zha+23] Zhengyun Zhao et al. "A large-scale dataset of patient summaries for retrieval-based clinical decision support systems." In: *Scientific data* 10 1 (2023), p. 909. URL: https://api.semanticscholar.org/CorpusID:266360591.

[Liv+24] Gill Livingston et al. "Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission". In: *The Lancet* 404.10452 (Aug. 2024), pp. 572–628. ISSN: 0140-6736. DOI: 10.1016/S0140-6736(24)01296-0. URL: https://doi.org/10.1016/S0140-6736(24)01296-0.

A PMC Dataset Selection

In order to select the dementia patients the following keywords were searched on the patients summaries in the PMC dataset:

```
DEMENTIA_KEYWORDS = [
"dementia", "Alzheimer", "Alzheimer’s", "vascular dementia",
"Lewy body", "frontotemporal dementia", "cognitive decline",
"memory loss", "mild cognitive impairment", "MCI"
]
```

For the control the following keywords were used:

```
CONTROL_KEYWORDS = [
"hip fracture", "hip replacement", "knee replacement",
"cataract", "macular degeneration", "osteoporosis",
"prostate cancer", "hernia repair", "pneumonia",
"urinary tract infection", "atrial fibrillation"
]
```

Then their dementia labels were checked and a random selection of 350 from each class was used.

B Hyperparameters

The following hyperparameters were used to train XGBoost models.

```
n_estimators=100,
max_depth=4,
random_state=42,
eval_metric=’logloss’
```